

Approximate Nearest neighbor problem amidst box obstacles in $\mathbb{R}^d$

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Input

- B : the set of m disjoint axis-aligned boxes in $\mathbb{R}^d$
- $S(= \{s_1, \dots, s_k\})$: the set of k sites in $\mathbb{R}^d$
- Query point $q \in \mathbb{R}^d$
- Approximation factor ε
- The number of all vertices of boxes and sites is $n := 2^d m + k$.

Output

- Return one of s_i such that $d(q, s_i) \leq (1 + \varepsilon) \cdot d(q, s_{i^*})$ where $i^* = \arg \min_i d(q, s_i)$ and $d(x, y)$ is the Euclidean geodesic distance from x to y .

Sketch of the algorithm

1. For any $(d-2)$ -face of the box $\Pi_{j=1}^d [a_{ij}, b_{ij}]$, generate Steiner points in the following way.
 - Without loss of generality, suppose that a $(d-2)$ -face is represented as $\Pi_{j=1}^{d-2} [a_{ij}, b_{ij}] \times a_{i(d-1)} \times a_{id}$.
 - The interval is defined as $\varepsilon' = \frac{r_{\min}}{(2d+1)^{\frac{3}{2}}} \varepsilon$ where $r_{\min}$ is the distance of the closest pair on $B \cup S \cup \{q\}$.
 - Then, we place Steiner points on each point in $\Pi_{j=1}^{d-2} \{a_{ij} + m_{ij} + l\varepsilon' \mid l = 0, 1, \dots, \lfloor \frac{b_{ij}-a_{ij}}{\varepsilon} \rfloor\} \times \{a_{i(d-1)}\} \times \{a_{id}\}$, where the j -th *margin* m_{ij} is defined as $\{(b_{ij} - a_{ij}) - \lfloor \frac{b_{ij}-a_{ij}}{\varepsilon'} \rfloor \cdot \varepsilon'\} / 2$.
2. Construct the visibility graph $G = (V_\varepsilon, E_\varepsilon)$ with respect to B where V_ε is the union of $S \cup \{q\}$ and the Steiner points generated in 1.
3. Find s_i that minimizes $d_G(q, s_i)$ by running Dijkstra algorithm on $G = (V_\varepsilon, E_\varepsilon)$.

Correctness of the algorithm

We show that $d_G(q, s_i) \leq (1 + \varepsilon)d(q, s_i)$ for any $s_i \in S$. Let $\pi = p_1 \dots p_{m'} (p_1 = q \text{ and } p_{m'} = s_i)$ be the shortest Euclidean geodesic path from q to s_i .

1. For any $1 < l < m'$, p_k is on a $d - 2$ -dimensional face of an obstacle. If p_l is on the interior of a $(d - 1)$ -dimensional face, $|p_{l-1}p_l| + |p_l p_{l+1}|$ can be reduced by moving p_l on the face.
2. For any $1 < l < m'$, there exists a Steiner point p'_i contained in $Q_l := \Pi_{i=1}^d [p_{ki} - \frac{\varepsilon'}{2}, p_{li} + \frac{\varepsilon'}{2}]$ where p_{li} is the i -th coordinate of p_l .
3. For $1 \leq l < m'$, the maximum distance from Q_l to Q_{l+1} is not more than $|p_{l-1}p_l| + \frac{\varepsilon}{2d+1} \cdot r_{\min}$.
4. Suppose that π hits the box b_1 and b_2 in B concecutively. Let $p_{l_1} \in \pi$ be the first point hitting b_1 and $p_{l_2} \in \pi$ be the first point hitting b_2 . Then the number of segments of π between p_{l_1} and p_{l_2} is at most $2d + 1$ since the number of $(d - 1)$ -dimensional faces of a box is $2d$ and each face contain at most one segment of π .
5. Therefore, $d_G(p_{l_1}, p_{l_2}) \leq \sum_{i=l_1}^{l_2-1} |p'_i p'_{i+1}| \leq d(p_{l_1}, p_{l_2}) + \varepsilon \cdot r_{\min} \leq (1 + \varepsilon) \cdot d(p_{l_1}, p_{l_2})$.

Complexity of the algorithm

Let denote $C := \left(\frac{l_{\max}}{r_{\min}}\right)^d$ where $l_{\max}$ is the longest side of the boxes in B . The number of faces of all boxes is $O(d^2 m) = O(n)$ since $d^2 \leq 2^d$ for all $d \geq 4$. Then, the number of Steiner points is $O\left(\frac{d^{1.5(d-2)}}{\varepsilon^{d-2}} n\right)$ and the number of edges in the visibility graph is $O\left(\frac{d^{3(d-2)}}{\varepsilon^{2(d-2)}} n^2\right)$. Also, we can determine that a segment intersects with B in $O(dm) = O(n)$ time. So, we can construct the visibility graph in $O\left(\frac{d^{3(d-2)}}{\varepsilon^{2(d-2)}} n^2 \left(n + \log\left(\frac{d}{\varepsilon} n\right)\right)\right)$ time since and the Dijkstra from the query point spends $O\left(\frac{d^{3(d-2)}}{\varepsilon^{2(d-2)}} n^2 \log\left(\frac{d}{\varepsilon} n\right)\right)$ time.